
```
//program to turn ON an LED for 1 second and Off for 2 seconds
#define LED_Pin 6
void setup()
{
    pinMode(LED_Pin,OUTPUT);
}
void loop()
{
    digitalWrite(LED_Pin,HIGH);
    delay(1000);
    digitalWrite(LED_Pin,LOW);
    delay(2000);
}
```

```
//program to turn ON/OFF an LED based sensor detection
#define LED_pin 6
#define IR_sensor 7
void setup()
{
    pinMode(LED_pin,OUTPUT);
    pinMode(IR_sensor,INPUT);
}

void loop()
{
    if(digitalRead(IR_sensor) == 1)
    {
        digitalWrite(LED_pin,HIGH);
    }
    else
    {
        digitalWrite(LED_pin,LOW);
    }
}
```

```
#include <DHT.h>

#define DHTPIN 5

#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

void setup()
{
    Serial.begin(9600); dht.begin();
}

void loop()
{
    float t = dht.readTemperature();
    Serial.print("Temperature = ");
    Serial.print(t, 2);
    Serial.println(" C");
    float h = dht.readHumidity();
    Serial.print("Humidity = ");
    Serial.print(h, 2);
    Serial.print(" %");
    delay(500);
}
```

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>
#define SCREEN_WIDTH 128 // OLED display width, in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels
#define OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
#define DHTPIN 5
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
void setup() {
  Serial.begin(9600);
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("SSD1306 allocation failed"));
    for (;;);
  }
  dht.begin();
}
void loop() {
  float t = dht.readTemperature();
  float h = dht.readHumidity();
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
  display.setCursor(0,0);
  display.print("Temperature: ");
  display.print(t);
  display.println(" C");
  display.print("Humidity: ");
  display.print(h);
  display.println(" %");
  display.display();
  delay(2000); // Wait 2 seconds before taking another reading
}
```

```
void setup()
{
    pinMode(3, INPUT); //push button to digital pin-3
    pinMode(4, OUTPUT); //digital pin-4 to relay input
}
void loop()
{
    if(digitalRead(3))
        digitalWrite(4, HIGH);
    else
        digitalWrite(4, LOW);
    delay(500);
}
```

```
#define SoilMoisturePin 5
void setup()
{
    pinMode(SoilMoisturePin, INPUT);
    Serial.begin(9600);
}
void loop()
{
    int soilState = digitalRead(SoilMoisturePin);
    if (soilState == LOW)
    {
        Serial.println("Soil is Wet");
    }
    else
    {
        Serial.println("Soil is Dry");
    }
    delay(1000);
}
```

```
-----
```

```
#define LDRPin 5
#define LEDPin 6
void setup()
{
    pinMode(LDRPin, INPUT);
    pinMode(LEDPin, OUTPUT);
    Serial.begin(9600);
}
void loop()
{
    if (digitalRead(LDRPin) == LOW)
    {
        digitalWrite(LEDPin,HIGH);
    }
    else
    {
        digitalWrite(LEDPin,LOW);
    }
    delay(1000);
}
```

PROGRAM:

```
#define TRIG_PIN 7
#define ECHO_PIN 6
void setup() {
    Serial.begin(9600);
    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);
}
void loop() {
    long duration;
    float distance_cm;

    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    duration = pulseIn(ECHO_PIN, HIGH);
    distance_cm = duration * 0.034 / 2;
    Serial.print("Distance: ");
    Serial.print(distance_cm);
    Serial.println(" cm");
    delay(500);
}
```



```

// DHT11
#include <DHT.h>
#include <ThingSpeak.h>
#include <ESP8266WiFi.h>

unsigned long ChannelNumber = 2455430; //replace channel number
const char *WriteapiKey = "PXWT5VHK2YFQ9XPM"; //Write API key
const char *ssid = "Basu"; // replace with your SSID
const char *pass = "Lalita@1234"; // Replace with your Password
const char *server = "api.thingspeak.com";

#define DHTPIN D5
DHT dht(DHTPIN, DHT11);
WiFiClient client;

void setup()
{
  Serial.begin(115200);
  delay(10);
  dht.begin();
  ThingSpeak.begin(client);
  Serial.println("Connecting to ");
  Serial.println(ssid);
  WiFi.begin(ssid, pass);
  while(WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.println("Wifi connecting.....");
  }
  Serial.println("Wifi connected successfully ");
}

void loop()
{
  float t = dht.readTemperature();
  float h = dht.readHumidity();
  ThingSpeak.writeField(ChannelNumber, 1, t, WriteapiKey);
  ThingSpeak.writeField(ChannelNumber, 2, h, WriteapiKey);
  delay(3000);
}

```

PROGRAM:

```
#include <ThingSpeak.h>
#include <ESP8266WiFi.h>

const char *ssid = "Basu";
const char *password = "Lalita@1234";
unsigned long ChannelNumber = 2455430; // enter your Channel ID
const char * ReadAPIKey = "EERAKJFY8530Z0PK"; //Read API Key
WiFiClient client; //need to communication with the ThingSpeak server

void setup()
{
    Serial.begin(9600);
    WiFi.begin(ssid, password);
    ThingSpeak.begin(client);
    while(WiFi.status() != WL_CONNECTED)
    {
        delay(500);
        Serial.println("Wifi connecting.....");
    }
    Serial.println("Wifi connected successfully ");
}

void loop()
{
    long t = ThingSpeak.readLongField(ChannelNumber,1,ReadAPIKey);
    long h = ThingSpeak.readLongField(ChannelNumber,2,ReadAPIKey);
    Serial.print("Temperature: ");
    Serial.print(t);
    Serial.println(" C");
    Serial.print("Humidity: ");
    Serial.print(h);
    Serial.println(" %");
    delay(3000);
}
```

```

#include <ESP8266WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>

// WiFi credentials
const char* ssid = "BASU";
const char* password = "Lalita12";

// Telegram Bot Token from BotFather
const char* botToken = "7837914639:AAGnWfOD4GF-idkQC061cqEERi3sVgJhKiM";
String chat_id = "7837914639";

WiFiClientSecure client;
UniversalTelegramBot bot(botToken, client);

const int ledPin = D1;

void setup()
{
  Serial.begin(115200);
  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, LOW);
  WiFi.begin(ssid, password);
  client.setInsecure(); // Ignore certificate check
  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }
  Serial.println("WiFi connected");
}

void loop()
{
  int newMsg = bot.getUpdates(bot.last_message_received + 1);
  if (newMsg)
  {
    String msg = bot.messages[0].text;
    if (msg == "/on")
    {
      digitalWrite(ledPin, HIGH);
    }
    else if (msg == "/off")
    {
      digitalWrite(ledPin, LOW);
    }
  }
}

```

```
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <DHT.h>
const char *ssid = "BASU";
const char *pass = "Lalita@708070";
const char *mqtt_server = "test.mosquitto.org";
WiFiClient espClient;
PubSubClient client(espClient);
#define DHTPIN D4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
void setup()
{
  Serial.begin(9600);
  WiFi.begin(ssid, pass);
  Serial.print("Connecting WiFi...");
  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }
  Serial.println("WiFi Connected successfully");
  client.setServer(mqtt_server, 1883);
  Serial.print("Connecting MQTT...");
  while (!client.connect("ESP8266Client1"))
  {
    Serial.print(".");
    delay(1000);
  }
  Serial.println("MQTT Connected successfully");
  dht.begin();
}
void loop()
{
  client.loop();
  float temp = dht.readTemperature();
  char msg[10];
  dtostrf(temp, 2, 2, msg);
  client.publish("esp8266/basavaraj/temperature", msg);
  Serial.print("Temp: ");
  Serial.println(msg);
  delay(2000);
}
```

PROGRAM 11

```
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
const char *ssid = "BASU";
const char *pass = "Lalita@708070";
const char *mqtt_server = "broker.hivemq.com";
WiFiClient espClient;
PubSubClient client(espClient);
void callback(char *topic,byte *payload,unsigned int length)
{
    Serial.println("Temperature received: ");
    for(int i=0;i<length;i++)
    {
        Serial.print((char)payload[i]);
    }
    Serial.println();
}
void setup()
{
    Serial.begin(115200);
    WiFi.begin(ssid,pass);
    Serial.println("WiFi connecting...");
    while(WiFi.status()!=WL_CONNECTED)
    {
        delay(500);
        Serial.println(".");
    }
    Serial.println("WiFi connected successfully");
    client.setServer(mqtt_server,1883);
    client.setCallback(callback);
    Serial.println("MQTT connecting...");
    while(!client.connect("ESP8266_Sub"))
    {
        delay(500);
        Serial.println(".");
    }
    Serial.println("MQTT connected successfully");
    client.subscribe("esp8266/basavaraj/temperature");
}
void loop()
{
    client.loop();
}
```